

Aim:

Exception Handling

Input a mobile number and check the given number is valid mobile number or not.

System Requirements:

Hardware

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage

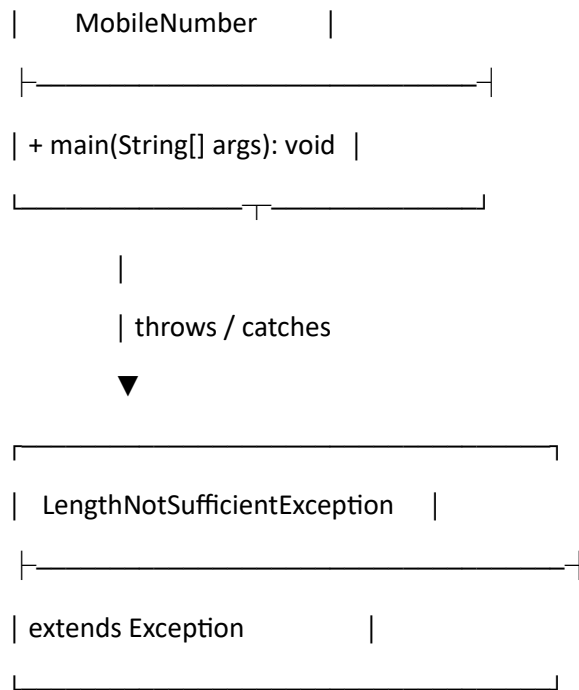
Software

- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

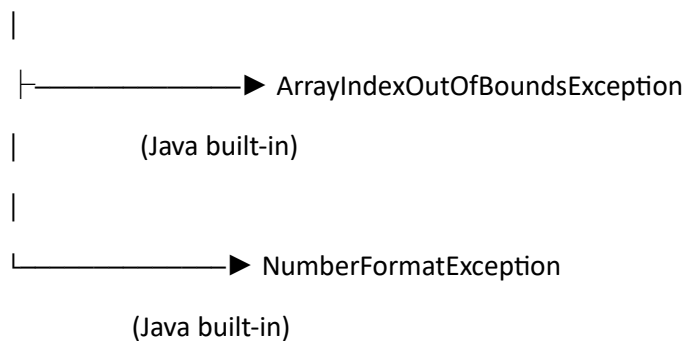
Algorithm:

1. Start the program.
2. Create a Scanner object to read input.
3. Ask the user to enter a mobile number.
4. Read the mobile number as a String.
5. Check whether the length is greater than 10.
6. If the length is greater than 10, throw `ArrayIndexOutOfBoundsException`.
7. Check whether the length is less than 10.
8. If the length is less than 10, throw `LengthNotSufficientException`.
9. Check each character of the mobile number.
10. If any character is not a digit, throw `NumberFormatException`.
11. If all conditions are satisfied, display "**Valid number**".
12. Handle the exceptions using separate catch blocks.
13. Stop the program.

UML Class Diagram:



MobileNumber



Code:

```

package javacore;

import java.util.Scanner;

class MobileNumber {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter mobile number: ");
    }
}
  
```

```

String n = sc.next();

try {
    if (n.length() > 10)
        throw new ArrayIndexOutOfBoundsException();

    if (n.length() < 10)
        throw new LengthNotSufficientException();

    for (int i = 0; i < n.length(); i++) {
        if (!Character.isDigit(n.charAt(i)))
            throw new NumberFormatException();
    }

    System.out.println("Valid number");
}
catch (ArrayIndexOutOfBoundsException e) {
    System.out.println("Invalid Mobile Number-ArrayIndexOutOfBoundsException");
}
catch (LengthNotSufficientException e) {
    System.out.println("Invalid Mobile Number-LengthNotSufficientException");
}
catch (NumberFormatException e) {
    System.out.println("Invalid Mobile Number-NumberFormatException");
}
}

class LengthNotSufficientException extends Exception {
}

```

Output:

Enter mobile number: 9885089465

Valid number

Enter mobile number: 98567890121

Invalid Mobile Number-ArrayIndexOutOfBoundsException

Enter mobile number: 88664433

Invalid Mobile Number-LengthNotSufficientException

Enter mobile number: 98ab@123

Invalid Mobile Number-NumberFormatException